

31. Morphological operations based on OpenCV using Opening technique.

AIM:

Do morphological operations in an image.

PROGRAM:

```
import cv2
import numpy as np

img = cv2.imread(r"C:\Users\Susmitha\Downloads\ITA0510 Lab\exp1.png")

if img is None:
    print("Error: Image not found!") else:
    # Create a 5x5 kernel
    kernel = np.ones((5, 5), np.uint8)

    # Apply Opening operation (Erosion followed by Dilation)
    opening = cv2.morphologyEx(img, cv2.MORPH_OPEN, kernel)

    # Save the output image
    cv2.imwrite(
        r"C:\Users\Susmitha\Downloads\ITA0510 Lab\Opened.jpg",
        opening
    )

    # Display the images
    cv2.imshow("Original Image", img)
    cv2.imshow("Opened Image", opening)

    cv2.waitKey(0)
    cv2.destroyAllWindows()
```

INPUT:



OUTPUT:



RESULT:

Thus the program is executed successfully.